A Numerical Algorithm for simulating multiphase flows: ice, melt, vapor, deformable solids, high speed, and low speed flows *

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June 14, 2026

ABSTRACT: A new Eulerian method is presented for simulating complex multiphase flows consisting of any number of types of materials, to include compressible fluids, incompressible fluids, ice, melt, vapor, and solids undergoing plastic deformation. The interfaces between materials is modeled and discretized as sharp.

*this material is based upon work supported by National Aeronautics and Space Administration under grant number 80NSSC20K0352

1 Mathematical Modeling

There are M materials and each material $1 \leq m \leq M$ occupies the region,

$$\Omega_m(t) = \{(\vec{x},t)|\phi_m(\vec{x},t) \geq 0\} \tag{1}$$

The $\phi_m(\boldsymbol{x},t)$ are level set functions[2, 1]. The zero level set of such functions demarcate the boundary of material m:

$$\partial\Omega_m(t) = \{(\vec{x},t))|\phi_m(\vec{x},t) = 0\} \tag{2}$$

In the bulk regions of material m we have the following equations for compressible materials:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{3}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p\mathbb{I} - \tau) = \vec{F}_{body} \tag{4}$$

$$(\rho E)_t + \nabla \cdot ((\rho E + p)\vec{u} - \vec{u} \cdot \tau - k\nabla T) = \vec{u} \cdot \vec{F}_{body} \tag{5}$$

$$(\rho Y)_t + \nabla \cdot (\rho Y \vec{u}) = \nabla \cdot (\rho D \nabla Y) \tag{6}$$

$$\tau = 2\mu \bar{D} \tag{7}$$

$$\bar{D} = D - \frac{\text{Trace}(D)\mathbb{I}}{\text{Trace}(\mathbb{I})} \tag{8}$$

$$D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \tag{9}$$

$$\rho E = \rho e_{\text{int}} + \frac{1}{2}\rho|\vec{u}|^2 \tag{10}$$

$$p = p(\rho, e) \quad e = c_v T \tag{11}$$

If the material is a viscoelastic FENE-CR material, then one has the additional equation for the configuration tensor A,

$$A_t + \vec{u} \cdot \nabla A = A \cdot (\nabla \vec{u})^T + \nabla \vec{u} \cdot A - \frac{f(R)}{\lambda}(A - \mathbb{I}), \tag{12}$$

and the momentum constitutive law term is modified,

$$\tau = \tau + \frac{\tilde{\eta}_P}{\lambda} f(R) A. \tag{13}$$

If the material is an elastic material undergoing plastic deformation, then one has the following additional equation for the deviatoric stress tensor,

$$s_t + \vec{u} \cdot \nabla s + s\Omega - \Omega s = 2G(\bar{D} - D^p), \tag{14}$$

and the momentum constitutive law term is modified,

$$\tau = \tau + s. \tag{15}$$

If the material is an elastic material undergoing plastic deformation, then the plastic strain equation is solved too,

$$\epsilon_t^p + \vec{u} \cdot \nabla \epsilon^p = \sqrt{\frac{2}{3}}|\Lambda|. \tag{16}$$

$$|\Lambda| = \sqrt{\frac{3}{2}}\sqrt{D^p : D^p} \tag{17}$$

$$D^p = \Lambda N \quad N = \sqrt{3/2}\frac{s}{S_e} \quad (\dot{\epsilon}^p)^2 = \frac{2}{3}\Lambda^2 \tag{18}$$

Ω is the spin tensor,

$$\Omega_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}) \tag{19}$$

The effective stress is defined as,

$$S_e = \sqrt{\frac{3}{2}}\sqrt{s : s} \tag{20}$$

The Johnson-Cook model yield stress is given by,

$$\sigma_y = (A + B(\epsilon^p)^n)(1 + C\ln(\frac{\dot{\epsilon}^p}{\dot{\epsilon}_0}))(1 - \theta^m) \quad \theta = \frac{T - T_0}{T_m - T_0} \tag{21}$$

One form of the Zerilli Armstrong model yield stress is given by,

$$\beta = \beta_0 - \beta_1 \log(\dot{\epsilon}^p / \dot{\epsilon}_{ref}^p) \tag{22}$$

$$\sigma_y = A + B(\epsilon^p)^n e^{-\beta T} \tag{23}$$

The energy contribution due to elasto-plastic deformation is modeled as,

$$\beta \dot{\epsilon}^p S_e, \tag{24}$$

and is added to (5). In order to determine Λ, as it appears in (18), the following "radial return algorithm" is implemented (see [3] for the case when $n = 1$ in (21) or (23)):

1. $\dot{\epsilon}^p = 0$, $k = 0$

2. $S_e = \sqrt{\frac{3}{2}}\sqrt{s : s}$

3. $\sigma_y = \sigma_y(T, \epsilon^p, \dot{\epsilon}^p)$

4. if $S_e - \sigma_y \le 0$ then $\Lambda = 0$ (i.e. $D^p : D^p = 0$).

5. if $S_e > \sigma_y$ then,

6. Solve the following equation for ϵ^p_{new}:

$$S_e - \sigma_y = 3G(\epsilon^p_{new} - \epsilon^p) + B((\epsilon^p_{new})^n - (\epsilon^p)^n) \tag{25}$$

7.

$$\dot{\epsilon}^p = \frac{\epsilon^p_{new} - \epsilon^p}{\Delta t} \tag{26}$$

8.

$$\Lambda = \sqrt{\frac{3}{2}}\dot{\epsilon}^p \tag{27}$$

$$D^p = \Lambda N \tag{28}$$

9. $k = k + 1$

10. if $k = 1$ then go back to step 3.

The mathematical model at sharp interfaces is given as follows. First, define a "closest point" function:

$$\vec{x}_c(\vec{x}) = \text{argmin}_{\vec{y},m}\left\{||\vec{y} - \vec{x}|||\phi_m(\vec{y}) = 0\right\} \tag{29}$$

Taking into account mass transfer, a source material's level set function (e.g. ice that is melting or liquid that is turning into vapor) satisfies

$$(\phi_{m_s})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_s} = -\frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}}|\nabla\phi_{m_s}| \tag{30}$$

The destination material's level set function satisfies

$$(\phi_{m_d})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}}|\nabla\phi_{m_d}| \tag{31}$$

The above equations are equivalent to the following:

$$(\phi_{m_s})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_s} = -\frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}}|\nabla\phi_{m_s}| \tag{32}$$

$$(\phi_{m_d})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}}|\nabla\phi_{m_d}| \tag{33}$$

Now, we define the interface normal:

$$\vec{n}_m = \frac{\nabla\phi_m}{||\nabla\phi_m||} \tag{34}$$

The interface curvature is,

$$\kappa_m = \nabla \cdot \vec{n}_m \tag{35}$$

The rate of mass transfer is,

$$\dot{m} = \frac{k_{m_s}\nabla T_{m_s} - k_{m_d}\nabla T_{m_d}}{L} \cdot \vec{n}_{m_d}. \tag{36}$$

The phase change interface temperature condition is,

$$T_{m_s} = T_{m_d} = T_{\text{saturation}}(P_{\text{equilibrium}}) \tag{37}$$

For evaporation, the following rate of mass transfer should match that in (36):

$$\dot{m} = \frac{-\vec{n}_{m_s} \cdot \nabla Y_{m_d}\rho_{m_d} D}{Y^{\Gamma} - 1} \tag{38}$$

Let the tensor σ represent pressure, viscosity, viscoelastic, and elastic contributions to the momentum equations. One has:

$$\sigma = -p\mathbb{I} + \tau. \tag{39}$$

The stress jump conditions are

$$\vec{n}_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = -\gamma_{m_s,m_d}\kappa_{m_s} - \dot{m}^2(\frac{1}{\rho_{m_s}} - \frac{1}{\rho_{m_d}}) \tag{40}$$

$$(\vec{t}_{i,tangent})_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = 0 \quad i = 1, 2 \tag{41}$$

The velocity jump conditions are

$$(\vec{u}_{m_s} - \vec{u}_{m_d}) \cdot \vec{n}_{m_s} = \dot{m}(\frac{1}{\rho_{m_d}} - \frac{1}{\rho_{m_s}}) \tag{42}$$

2 phase change algorithm

See the following articles and below: [5], [4], [6].

1. For each cell in a narrow band about the interface, use the closest point map to find the corresponding closest point on the interface. Approximate the phase change speed, $|\dot{m}|/\rho$ at the closest point.

2. extend the phase change speed $|\dot{m}|/\rho$ further into bulk regions.

3. If distributing the mass source term due to expansion or compression, $\dot{m}$, to the destination material side, then the velocity on the interface corresponds to the source material velocity and the overall interface velocity is:

$$\vec{u}_{source} + \frac{|\dot{m}|}{\rho_s}\vec{n}_s \tag{43}$$

If distributing $\dot{m}$ to the source material side, then the velocity on the interface corresponds to the destination material velocity and the interface velocity is:

$$\vec{u}_{dest} + \frac{|\dot{m}|}{\rho_d}\vec{n}_s \tag{44}$$

4. Using step 3 as a guide, do unsplit CISL phase change advection. If distributing $\dot{m}$ to the destination material side, the phase change contribution to the interface velocity is

$$\frac{|\dot{m}|}{\rho_s}\vec{n}_s \tag{45}$$

5. The difference in volume fraction from step 4 gives the $\dot{m}^{conservative}$ expansion source term to be applied during the pressure update step.

6. The $\dot{m}^{conservative}$ terms are distributed to one side of the interface. If distributed to the destination material side, then distribute where $F_{dest} > 1/2$. This guarantees, for linear interfaces, that all faces with at least one adjoining source material majority cell will have the source material velocity defined at that face. Please refer to Figure 1.

For compressible flow, with a predictor and a corrector step, there is a problem. For the evaporating droplet problem, the $\dot{m}$ terms are distributed to the vapor (destination). Please refer to Figure 1. If the $\dot{m}$ terms are not distributed deep enough into the bulk of the destination material, then:

1. During the predictor step, there is no problem because the advection velocity at time t^n is used in order to advect the interface that is initially defined at $t = t^n$ (Γ^n).

2. During the corrector step, there is a problem because the advection velocity at time t^{n+1} is used in order to advect the interface that is initially defined at $t = t^n$ (Γ^n). The problem is that some of the $\dot{m}$ terms distributed to the vapor side of the Γ^{n+1} interface will be on the liquid side of the $t = t^n$ (Γ^n) interface. See the blue region of Figure 1.

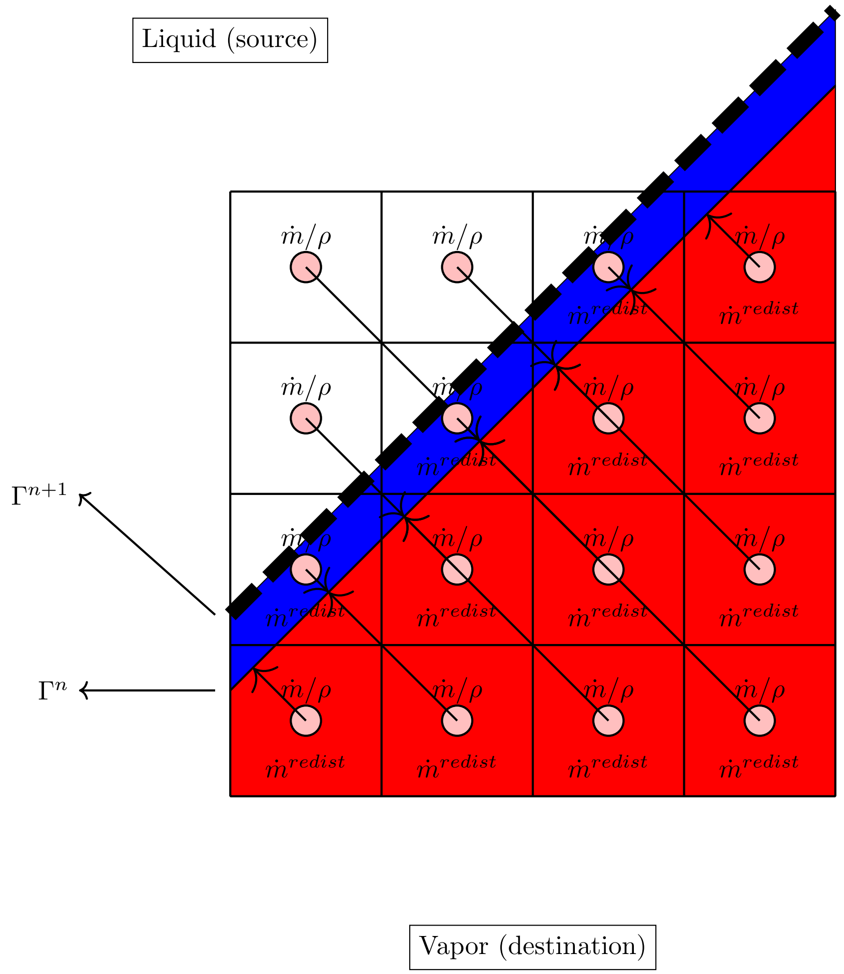

Figure 1: The phase change algorithm: (1) the phase change speed, $\dot{m}/\rho$, is approximated at all narrow band cells by evaluating $\dot{m}/\rho$ at the corresponding closest point on the interface (Γ^n). (2) The change in volume fraction due to unsplit CISL advection determines the mass source, $\dot{m}$, for the projection step. (3) the mass source, $\dot{m}$ is distributed to the vapor side of the Γ^{n+1} interface. At the next time step, the interface is advected using $\vec{u}^{source}$. For compressible flow, there is a problem if $\dot{m}$ is not distributed deep enough into the vapor material. This is because the corrector advection of Γ^n will now have some redistributed $\dot{m}$ terms on the liquid side of the Γ_n interface (the blue region).

3 Applications

1. Jet in cross flow problem (military jet engines, Pratt and Whitney)

2. Diesel injectors

3. Rotating Detonation Engines (see latest article with Everett Wenzel and Marco Arienti).

4 Algorithm

1. Cell integrated semi Lagrangian advection

2. Cell integrated semi Lagrangian advection due to phase change.

3. Viscosity, Viscoelastic, and elastic forces.

4. Temperature Diffusion

5. Mass Fraction Diffusion

6. Pressure Gradient

7. Go back to step 1.

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